

# Zepto Inventory & Pricing Analysis Report

## Business Problem

Zepto is a quick-commerce grocery delivery platform that lists thousands of SKUs across dozens of categories, each with its own MRP, discount, stock level, and package weight. Effective catalog and inventory management is critical to staying price-competitive while protecting margins. This analysis uses SQL to explore the Zepto product catalog and answer the following business questions:

- Which products offer customers the best discount-based value?
- Which high-value products are frequently unavailable, representing lost revenue?
- Which categories generate the most estimated revenue?
- Which premium products are under-discounted relative to their price point?
- Which categories are most aggressive with discounting?
- Which products offer the best price-per-gram value?
- How is inventory distributed across weight/package-size tiers?
- Which categories carry the heaviest total inventory load?

**Tech stack:** PostgreSQL (data exploration, cleaning, and analysis) — see `zepto_project.sql` for the full query set.

## Exploratory Data Analysis Insights

### Dataset Overview

| Metric                                                          | Value         |
|-----------------------------------------------------------------|---------------|
| Total rows (raw)                                                | 3,732         |
| Total rows (after cleaning)                                     | 3,731         |
| Categories                                                      | 14            |
| Distinct product names appearing more than once (multiple SKUs) | 1,214         |
| Out-of-stock SKUs                                               | 453 (12.1%)   |
| In-stock SKUs                                                   | 3,278 (87.9%) |
| Null values found                                               | None          |

Summary Statistics (post-cleaning)

| Column                       | Mean   | Std Dev | Min   | 25%   | Median | 75%    | Max      |
|------------------------------|--------|---------|-------|-------|--------|--------|----------|
| MRP (₹)                      | 156.84 | 160.89  | 10.00 | 60.00 | 110.00 | 200.00 | 2,600.00 |
| Discount %                   | 7.62   | 9.21    | 0     | 0     | 6      | 10     | 51       |
| Available Quantity           | 4.01   | 2.20    | 0     | 2     | 5      | 6      | 6        |
| Discounted Selling Price (₹) | 141.97 | 138.51  | 9.00  | 55.00 | 104.00 | 184.00 | 1,399.00 |
| Weight (g)                   | 387.93 | 678.17  | 0     | 100   | 225    | 450    | 10,000   |

Data Quality Notes

**Zero-price row:** One SKU (`sku_id 3607`) had both MRP and discounted selling price of 0 — an invalid listing, likely a data entry error. It was removed before analysis.

**Currency scale:** Prices in the raw table were stored in paise (e.g., 2500 = ₹25.00). All `mrp` and `discountedsellingprice` values were divided by 100 to convert to rupees before analysis.

**Repeated product names:** 1,214 product names appear more than once in the catalog (different SKUs for different pack sizes/weights of the same product — e.g., "Arden Eggs White" and "Amul Pure Ghee (Pouch)" each appear in up to 10 variants). This is expected for a grocery catalog and was treated as normal, not a duplicate/error.

**Duplicate category-level totals:** Several category pairs (Dairy, Bread & Batter / Beverages; Chocolates & Candies / Packaged Food / Ice Cream & Desserts; Personal Care / Paan Corner; Cooking Essentials / Munchies) share identical row counts and, as a result, identical aggregate revenue and weight totals in the queries below. This appears to be a structural artifact of how the source dataset was assembled rather than a coincidence, and is worth flagging if this catalog is used for further modeling.

**Discount distribution is right-skewed:** 25% of products carry no discount at all, while a small number of items (mostly wafers, cheeses, and yogurts nearing their category ceiling) are discounted as much as 51%.

Correlation Insights

|            | MRP  | Discount % | Avail. Qty | Selling Price | Weight | Order Qty |
|------------|------|------------|------------|---------------|--------|-----------|
| MRP        | 1.00 | 0.17       | 0.11       | 0.97          | 0.44   | 0.08      |
| Discount % | 0.17 | 1.00       | 0.09       | 0.04          | 0.01   | 0.02      |
| Weight (g) | 0.44 | 0.01       | 0.04       | 0.47          | 1.00   | 0.01      |

- **MRP vs. Selling Price:** Very strong correlation (0.97) — as expected, since selling price is derived directly from MRP and discount.
  - **MRP vs. Discount %:** Weak positive correlation (0.17), suggesting pricier products get *slightly* deeper discounts on average, but price is not a strong driver of discount depth.
  - **Weight vs. Discount %:** Essentially no correlation (0.01) — heavier/bulkier products are not discounted any differently than smaller ones.
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## Research Questions & Key Findings

### 1. Best-Value Products by Discount Percentage

The top discounts in the catalog cap out at **51%**, led by Dukes Waffy Wafers (Chocolate/Orange/Strawberry, ₹45 MRP) and several RRO cheese and Epigamia yogurt products (all in the 50% range). These are almost entirely short-shelf-life dairy and snack items — consistent with retailers discounting perishables or slow-moving small-format snacks most aggressively.

### 2. High-MRP Products That Are Out of Stock

**4 products** priced above ₹300 are currently out of stock: Patanjali Cow's Ghee (₹565), MamyPoko Pants Diapers XL (₹399), Aashirvaad Atta with Multigrains (₹315), and Everest Kashmiri Lal Chilli Powder (₹310). While a small list, these are all high-repeat-purchase staples — their unavailability represents disproportionate lost revenue relative to catalog share.

### 3. Estimated Revenue by Category

Estimated revenue (discounted price × available quantity) ranges from **₹10,846** (Fruits & Vegetables — lowest) up to **₹337,369** (Cooking Essentials / Munchies — highest, tied). Bulky pantry staples and snacking categories dominate revenue contribution, while fresh produce, despite driving footfall, contributes the least — likely due to low per-unit prices and limited available quantity (capped at 3–6 units per SKU in this catalog).

| Category              | Est. Revenue (₹) |
|-----------------------|------------------|
| Fruits & Vegetables   | 10,846           |
| Meats, Fish & Eggs    | 20,693           |
| Biscuits              | 25,008           |
| Dairy, Bread & Batter | 55,051           |
| Beverages             | 55,051           |
| Health & Hygiene      | 64,180           |
| Home & Cleaning       | 122,661          |
| Chocolates & Candies  | 224,385          |
| Packaged Food         | 224,385          |
| Ice Cream & Desserts  | 224,385          |
| Personal Care         | 270,849          |
| Paan Corner           | 270,849          |
| Cooking Essentials    | 337,369          |
| Munchies              | 337,369          |

#### 4. Premium Products with Minimal Discounting

**39 products** have an MRP above ₹500 but a discount under 10% — mostly cooking oils in bulk jars (Dhara, Saffola, Fortune), infant formula, and premium hair color/shampoo brands. These are largely non-discretionary or brand-loyal purchases (cooking oil, baby formula) where price sensitivity is lower, which likely explains why they're held near full price.

#### 5. Categories With the Highest Average Discounts

| Category             | Avg. Discount % |
|----------------------|-----------------|
| Fruits & Vegetables  | 15.46           |
| Meats, Fish & Eggs   | 11.03           |
| Ice Cream & Desserts | 8.32            |
| Packaged Food        | 8.32            |

| Category             | Avg. Discount % |
|----------------------|-----------------|
| Chocolates & Candies | 8.32            |

Fresh, perishable categories (produce, meat/fish/eggs) carry the deepest average discounts — consistent with retailers moving short-shelf-life inventory quickly to avoid spoilage losses.

### 6. Best Price-Per-Gram Value (products ≥ 100g)

Vicks Cough Drops (₹0.017/g), Aashirvaad Iodised Salt (₹0.019/g), and bulk Onion/Potato/Tata Salt packs (₹0.019-0.029/g) offer the best value density in the catalog. Unsurprisingly, commodity staples sold in bulk (salt, onions, potatoes) dominate the best-value list — these are the categories where per-gram cost should anchor customer price perception.

### 7. Inventory by Weight Tier

| Weight Tier | Definition      | # of Distinct Products |
|-------------|-----------------|------------------------|
| Low         | < 1,000g        | 1,613                  |
| Medium      | 1,000g - 4,999g | 147                    |
| Bulk        | ≥ 5,000g        | 22                     |

The catalog is overwhelmingly composed of small-format items (91% under 1kg), reflecting Zepto's quick-commerce model of small, fast-delivery basket sizes rather than bulk/wholesale purchasing.

### 8. Total Inventory Weight by Category

| Category              | Total Weight (g) |
|-----------------------|------------------|
| Meats, Fish & Eggs    | 48,016           |
| Biscuits              | 84,431           |
| Fruits & Vegetables   | 91,794           |
| Health & Hygiene      | 142,904          |
| Dairy, Bread & Batter | 143,735          |
| Beverages             | 143,735          |
| Paan Corner           | 348,187          |

| Category             | Total Weight (g) |
|----------------------|------------------|
| Personal Care        | 348,187          |
| Home & Cleaning      | 373,161          |
| Chocolates & Candies | 490,797          |
| Packaged Food        | 490,797          |
| Ice Cream & Desserts | 490,797          |
| Cooking Essentials   | 1,404,654        |
| Munchies             | 1,404,654        |

Cooking Essentials and Munchies carry the heaviest total inventory load by a wide margin — consistent with them also being the top revenue-generating categories, suggesting warehouse/storage capacity planning should prioritize these two categories.

### Final Recommendations

- **Restock high-MRP staples immediately** — ghee, atta, and spice SKUs going out of stock cost disproportionate revenue relative to their small count; prioritize replenishment alerts for products above ₹300 MRP.
- **Deepen discounts on perishables strategically** — Fruits & Vegetables and Meats/Fish/Eggs already carry the highest average discounts; monitor sell-through to confirm this is preventing spoilage losses rather than eroding margin unnecessarily.
- **Reassess pricing on low-discount premium staples** — cooking oils, formula, and hair color brands sit near full price; test targeted promotions to see if elasticity exists without damaging brand-loyalty pricing.
- **Use price-per-gram as a value benchmark** — commodity categories (salt, onions, potatoes) are the best "value" anchors in the catalog; feature them in value-focused marketing to build basket trust.
- **Right-size warehouse allocation** — Cooking Essentials and Munchies dominate both total inventory weight and revenue; ensure storage/logistics capacity scales with these two categories first.
- **Investigate the duplicated category totals** flagged in the data-quality notes before using this dataset for further quantitative modeling, to confirm whether it reflects real catalog structure or an upstream data assembly issue.

